

Review

The Applications and Challenges of Nanofluids as Coolants in Data Centers: A Review

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Abstract: With the rapid development of artificial intelligence, cloud computing and other technologies, data centers have become vital facilities. In the construction and operation of data centers, how to effectively solve the problem of cooling and energy saving is the key problem. In this review article, a critical review of recent research regarding the application of nanofluids in data center cooling are put forward. Many different aspects of nanofluids such as the classification of nanoparticles, base fluid components, and types and structures of heat exchangers were discussed. Furthermore, some advanced and up-to-date apparatus and theoretical models of utilizing nanofluids as coolants in data centers are reviewed and described in detail. Lastly, but not least, potential research directions in the future and the challenges faced by the researchers and industry in this field are proposed and discussed. In conclusion, nanofluids used as novel heat exchange medium, which has been widely proven in other areas, can also conspicuously improve data center cooling technology in the future.

Keywords: nanofluids; data centers; liquid cooling; coolants; heat transfer; applications; challenges



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1. Introduction

Data centers are one of the most important infrastructures of the information technology (IT) industry, such as those for 5G communication, cloud computing, Internet of things, algorithm development, and artificial intelligence, etc. [1]. Compared to the past, the rapid development of the current information technology industry has placed greater demands on the uninterrupted operation of key equipment in data centers [2]. It is predicted that data centers account for almost 2% of total global power consumption [3]. In recent years, China's information technology industry has developed rapidly, and has kept pace with developed countries in Europe and America in the fields of 5G communication and artificial intelligence. The rapid development of emerging frontier technologies such as supercomputer centers with high-performance computing capacity [3] and the Internet economy has brought a rapid increase in the demand for communication and computing [4]. For example, China's data centers used about 36 billion kilowatt hours of electricity in 2019, accounting for 1% of China's total electricity consumption. Since 2011, the total electricity consumption of data centers in China has grown rapidly at a rate of over 10% per year. By 2022, it reached more than 270 billion kilowatt hours, accounting for 2.71% of the total electricity consumption in society [3].

However, data centers consume large quantities of energy. As early as 2018, the total electricity consumption of global data centers had exceeded two hundred TWh [5]. In 2020, the electricity usage of data centers in the U.S. alone reached 70 TWh [6]. Compared to the standard office facilities, data centers expend up to one hundred times more electricity [7]. Such a huge energy demand poses a great challenge to the tense energy situation all over the world. According to research by Rong et al. [8], the electricity usage of refrigeration systems contributes approximately 40% of the total electricity consumption of data centers. The electricity consumption condition in data centers can be seen in Figure 1.

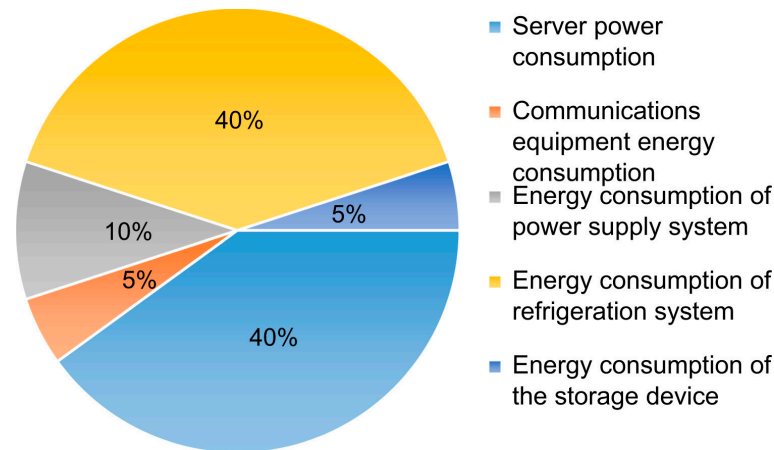


Figure 1. Electricity consumption composition of data centers. Reprinted from [8], Copyright (2016), with permission from Elsevier.

In the operation of traditional data centers, the main way to remove heat from high-power servers is air cooling [2]. However, the existing data center cooling technology is still based on air cooling [9]. Air cooling uses air as the cooling medium, and the convection heat transfer resistance of air is large while the heat carrying capacity is low. With the continuous improvement of chip heat flux density, air cooling technology is gradually becoming unable to satisfy the cooling needs of high heat flux microchips, and the phenomenon of IT equipment overheating and downtime occurs from time to time. High-performance chips pose a great challenge to existing air cooling technology, and the cooling problem has become a common problem faced by the industry.

Due to the high heat removal capacity of liquid cooled data centers, the use of liquid cooled methods in data centers has become increasingly popular in recent years [2]. Greenberg et al. [10] pointed out that using liquid cooling systems can substantially reduce the electricity consumption of refrigeration systems in data centers [10]. The difference between conventional air cooling and liquid cooling can be seen in Figure 2. In summary, using liquid cooled servers or racks, along with water/air side energy-saving devices with ambient natural air cooling, can reduce the use of CRAC devices and chillers, and in some cases, even eliminate such devices, thereby saving a lot of energy.

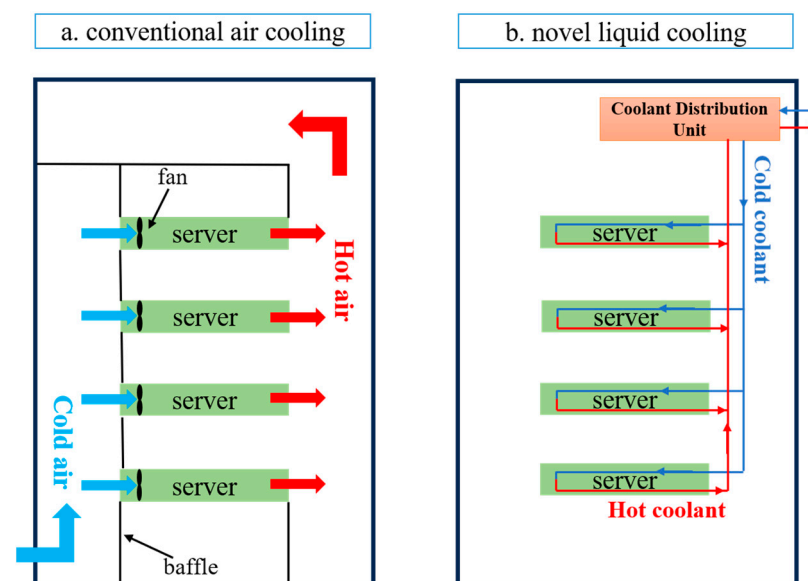


Figure 2. Schematic diagram of conventional air cooling and liquid cooling.

Due to the direct application of liquid in refrigeration systems, the supply and return water temperature of the liquid cooling system is likely to be higher, which enables liquid cooling data centers to have the capability for waste heat recirculation and energy capture [11]. The thermal resistance of liquid cooled data centers is 20% lower than data centers cooled by air, and the heat removal capacity exceeds 200 W/cm^2 [4].

Nanofluids are generally defined as fluids with suspended nanoparticles with an average diameter no more than 100 nm. It has been found that even a little number of nanosized particles added into the base fluid can improve the thermal transfer efficiency more than a pure fluid [12]. Nanofluids are utilized to replace the conventional fluid in systems which require heat transfer to be more efficient, especially in micro-sized applications [13].

For example, the employment of nanofluids in evacuated tube solar collectors was summarized in a review paper [12], which also investigated the thermal enhancement of nanofluid flow in a convergent–divergent loop [14] and a pipe with an innovative vortex generator [15]. The researchers utilized hybrid nanofluids within a parabolic trough solar concentrator equipped with inner helical axial fins to act as turbulators and, with the help of nanofluids, the heat transfer efficiency in the solar concentrator was enhanced compared to pure water [16].

The effect of nanofluids on the heat transfer in a tube with wall corrugations and center-cleared twisted-tape inserts was also studied. The performance of several nanofluids with water as a base fluid, including Al_2O_3 –water, CuO –water, copper–water, and TiO_2 –water, was investigated. The results showed that compared with pure water, nanofluids can improve the heat transfer rate and lead to a higher thermal efficiency, despite any associated rises in pressure drop [17].

The influence of water-based nanofluids on the heat transport in a shell and corrugated coil tube heat exchanger was also studied and the findings indicated that a CuO –water nanofluid exhibits the most superior thermal performance. However, the thermal performance decreased as the volume fraction of the nanofluid increased [18].

It has been observed that nanofluids can also be utilized in cold energy storage devices to align with thermal energy production and demand. The incorporation of a plate fin network and an MWCNT can accelerate the rate of ice formation by up to 70.14% [19].

The latest research by Shchegolkov et al. [20] shows that adding CNTs with a concentration of 1–7% to an organosilicon compound can increase the thermal conductivity from $0.2 \text{ W/(m}\cdot^\circ\text{C)}$ to $0.32 \text{ W/(m}\cdot^\circ\text{C)}$, an increase of up to 60%.

Various kinds of nanofluids have been used in the field of data center cooling [21–28]. Al_2O_3 –water nanofluids are widely utilized for data center cooling and are considered one of the most commonly used nanofluids in this application. Bahiraei et al. [29] designed a novel liquid block for Al_2O_3 –water nanofluids and measured the thermal and fluid properties. Many of the effects on the cooling efficiency of electronics, such as the Reynolds number and concentration, were studied and it was found that the effect of volume fraction and particle size on temperature is more significant than power.

The study also investigated the rheological characteristics and heat transfer efficiency of nanofluids consisting of graphene and deionized water. Vishnuprasad et al. [30] optimized a cooling system through the use of a multi-objective optimization approach and decision-based optimization method. Their results showed that compared with conventional coolants, the novel nanofluids have a significant influence on revising the thermofluid properties for improved performance.

Magnetic nanofluids can also be used for cooling electronic devices in data centers. In open-loop pulsating heat pipes, the reinforcement of heat transfer of magnetic nanofluids was investigated by Yunus et al. [31]. They found that magnetic nanofluids can be used as an effective thermal management technique due to their outstanding ability to transfer heat, which makes them ideal for use in electronic cooling systems.

The refrigeration characteristics of water-based CuO nanofluids in an electronic system were investigated by Sarafraz et al. [32] and compared with liquid gallium and water. The

CuO–water nanofluids were discovered to possess not just a greater thermal conductivity compared to water, but also a reduced requirement for pump delivery power.

To face the challenges of high heat flux on the integrated circuit chip cooling, liquid cooling technology is one of the most feasible technical solutions. Liquid cooling uses the cooling liquid (coolant) to take heat away from the high temperature chip surface directly for heat exchange. The liquid-to-chip heat transfer coefficient is much higher than that of air cooling, and the cooling capacity of the liquid is also significantly greater than that of air.

Therefore, liquid cooling technology plays a crucial role in addressing the cooling challenges posed by high heat flux integrated circuit chips. Liquid cooling technology represents the direction of the advanced technology in a new generation of IT equipment cooling development. It is the reason why many studies have been conducted to develop liquid cooling technology for data centers, which is crucial to promote the development of the new generation of information technology [33].

Previous literature has mostly focused on the application of nanofluids in traditional heat exchangers, and is lacking in coverage of the application of nanofluids in the field of liquid cooling in data centers. Secondly, most existing literature has studied the parameters of nanofluids, however, few have compared the performance of different nanofluids used in liquid cooling in data centers. In addition, there are a lack of reviews and comparisons of existing technologies regarding the application of nanofluids in novel liquid cooling technologies, such as phase change cooling. The goal of this article is to fill these gaps and provide a comprehensive review of nanofluids in liquid cooling systems in data centers. This article not only reviews various nanofluids on a laboratory scale, but also analyzes their applications in practical liquid cooling devices. This article not only reviews experimental research, but also summarizes numerical simulation and heat transfer and rheological models. In the second half of the article, based on the literature, this article analyzes the challenges of nanofluids in data center liquid cooling and provides suggestions for future research directions.

In this paper, the recent studies related to the utilization of nanofluids in data center cooling have been sorted. Various aspects of nanofluids, such as nanoparticle type, base fluid material, and types and structures of heat exchangers, are discussed. Moreover, some advanced and up-to-date apparatus that utilize nanofluids in the refrigeration of data centers are proposed and described in detail. Lastly, but not least, potential research directions for the future and the challenges faced by researchers and industry in this field are proposed and discussed.

The primary objective of the paper is to review the materials, methods, and research status of nanofluids for liquid cooling of data centers in recent years. The literature is classified by types of nanomaterials and application devices. Furthermore, the impact of various nanofluids on the thermal conductive properties and flow resistance of liquid cooling systems in data centers is discussed in detail. Finally, the problems of using nanofluids in the refrigeration process for data centers are pointed out and future development prospects and potential research directions are given.

2. Types of Nanofluids Used in Liquid Cooling

From Figure 3, we can see a lot of different nanofluids have been investigated in the application of liquid cooling. The nanoparticles used include metal, metal oxide, and carbon materials. The underlying spreading and splashing mechanisms of nanofluids exhibit significant differences compared to those of base fluids. After the incorporation of nanoscale particles, the characteristics of the base fluid are altered, as well as the interactions between particles, which is important to the movement of nanofluid. Many factors must be taken into consideration, such as thermal conductivity, insulation, viscosity, toxicity, and environmental impacts, etc. At present, it has been confirmed in the literature that carbon materials have excellent thermal physical parameters. Multi-walled carbon nanotubes can improve the thermal release performance of materials [34], and can be dispersed in fillers

to form conductive grids, regulating the electrical properties of materials [35]. However, metal oxide nanoparticles have been widely studied due to their lower cost.

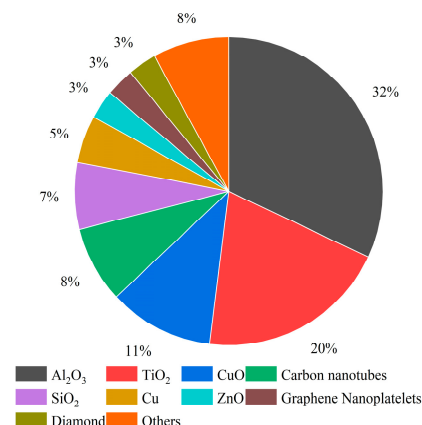


Figure 3. Percentage of researchers used specified nanoparticles in their research work.

2.1. Metal Nanoparticles

The most commonly utilized metal nanoparticles in the literature are those made of copper (Cu). Leong [36] conducted a study of improving heat transfer using ethylene glycol-based copper nanofluids in an automotive cooling system, testing different volume concentrations of nanoparticles, ranging from 0% to 2%, and analyzing the heat transfer rate at various Reynolds numbers for air and coolants. The findings indicated that increasing the concentration of nanoparticles led to a maximum heat transfer rate enhancement of 3.8%. Additionally, it was observed that higher Reynolds numbers for air resulted in a significant (45.2%) improvement in heat transfer rate when increased from 4000 to 6000. However, increasing the Reynolds number for the coolant from 5000 to 7000 only resulted in a minimal enhancement of heat transfer rate (0.4%). The main reason why the metal nanofluid with ethylene glycol as the base liquid increases the heat transfer rate by 3.8% is that the copper nanoparticles have a higher thermal conductivity compared to the base liquid, but the increase of 3.8% is smaller than those of water-based metal nanofluids. According to Nishant's explanation [37], the main reason is that water and ethylene glycol have different viscosities. The viscosity of ethylene glycol is higher than that of water, which limits the movement and interaction rate of nanoparticles, leading to a weakening of their Brownian motion.

Wadd et al. [38] dispersed copper nanoparticles (20 nm) into water and used sodium lauryl sulfate (SLS) as a surfactant to synthesize stable Cu–water nanofluids. After that, experiments were conducted in an automobile radiator to compare the thermal conductivity and fluid dynamics of copper and TiO₂ nanoparticle suspensions. The findings indicate that the thermal conductivity of Cu–water nanofluids surpasses that of TiO₂ nanofluids, which can be attributed to the superior thermal conductivity of copper nanoparticles, but Cu–water nanofluids have weaker stability than that of TiO₂ due to the copper nanoparticles' larger density.

Sheikhzadeh et al. [39] dispersed Cu nanoparticles (0–5% volume fraction) in ethylene glycol and observed that the Cu–ethylene glycol nanofluids can reinforce the overall coefficient for heat transfer up to 64.3%, and the heat transfer rate is as high as 29.6% in the air side. They also observed the overall coefficient for heat transfer on the air side for Cu–ethylene glycol nanofluids would increase by 12.4% if the Reynolds number was increased from 4000 to 6000. They also found that due to the addition of copper nanoparticles into working fluids, the radiator had a better thermal performance at high temperatures than that at low temperatures. That is, with the help of nanofluids, the temperature of the outgoing coolant fluid from radiator can be reduced. Their research results also showed that as the Reynolds number increases from 4000 to 7000, the heat transfer rate also increases. They attributed this to the increasing Reynolds number leading

to an increase in mass flow rate and the volumetric flow rate of air. Although the increase in the Reynolds number leads to a lower Colburn factor, the effect of mass rate is more pronounced. The heat transfer coefficient increases with the increase in mass flow rate. However, as the Reynolds number further increases, the percentage increase in heat transfer rate becomes less significant.

Another metal nanoparticle that has been utilized is silver (Ag). Behrangzade and Heyhat [40] conducted a comparison of the impact of utilizing a nanosilver–water nanofluid versus pure water on enhancing energy efficiency at the same flow rate and Reynolds number, respectively. They found that by using a 100 ppm nanosilver dispersed water-based nanofluid, there was a 16.79% improvement in the overall heat transfer coefficient, with no significant change in the pressure drop value.

Gold nanofluids were also used as refrigerants used in heat pipes for CPU cooling. Tsai et al. [41] added gold nanoparticles with different sizes into an aqueous solution. They found that compared with pure water, the Au nanofluids can reduce thermal resistance dramatically. Their study also suggested that different sizes of gold nanoparticles can result in different thermal resistance.

2.2. Metal Oxide Nanoparticles

Al_2O_3 nanofluids have been extensively investigated in numerous studies. Various volume fractions of Al_2O_3 nanoparticles were used to synthesise nanofluids in a study by Hassani et al. [42]. The Al_2O_3 nanofluids were utilized as a coolant for an electronic heat sink. The findings revealed that the addition of a 0.5% and 1.0% volume fraction of Al_2O_3 nanofluids can result in an average improvement of 14.7% and 28.3% in the performance index of the heat sink, respectively.

An Al_2O_3 –water nanofluid was studied by Nguyen [43] in a data center liquid cooling system. Within the various concentration levels and volumes of the Al_2O_3 –water nanofluid tested in their experiments, it was found that 6.8% of Al_2O_3 nanoparticles can increase the heat transfer coefficient by 40% compared to base fluid, and the size of the particles also impacts heat transfer, therefore a 36 nm Al_2O_3 particle is better than a 47 nm in terms of enhancing heat transfer coefficients. From Figure 4 we can see the heated block average temperature significantly dropped as the concentration of particles in the volume increased.

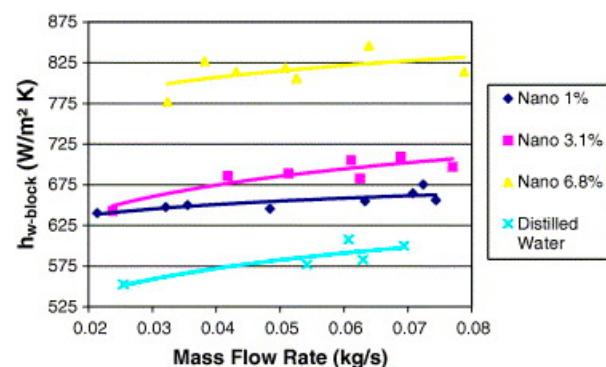


Figure 4. Effect of mass flow rate and particle concentration on convective heat transfer coefficient in water block. Reprinted from [43], Copyright (2007), with permission from Elsevier.

Nnanna [44] applied a water-based 27 nm Al_2O_3 nanofluid in a thermoelectric module with a heat source replicating a CPU and found that the thermoelectric module using the nanofluid had higher thermal efficiency. For example, the nanofluid can lead to a much lower temperature gradient than pure water. The lower the difference in temperature across the heat-conductive compound and the device for transferring heat is, the lower the resistance is when the heat flows across the thermal paste.

TiO_2 nanofluids can also be applied for data center cooling. Ambreen and Kim [45] simulated a 30nm spherical-shaped TiO_2 with a volume concentration of 4.31%. They made use of discrete phase model (DPM) to evaluate the combined effect of the TiO_2 nanofluids

and three distinct fin cross-sectional profiles on enhancing heat transfer efficiency. The Nusselt number, convective heat transfer coefficient distribution, the velocity streamlines and contours, and the temperature of the micro pin fin heat sink's base and surface were evaluated. Their results demonstrated that TiO_2 nanofluids perform much better than pure water and the circular fins cooled by the TiO_2 nanofluids have higher efficiency than the other fin cross-sectional shapes (hexagon and square). Compared to pure water, the addition of a 4.31% TiO_2 nanoparticle can bring 26%, 44%, and 62% raises in the Nusselt numbers of the heat sinks equipped with fins in square, circular, and hexagonal shapes, respectively.

Nitiapiruk [46] conducted a set of experiments to study the performance of a water-based TiO_2 nanofluid in a microchannel heat sink. By calculating the thermal conductive and viscosity properties of nanofluids consisting of TiO_2 and water through several different models, it was found that the most beneficial scenario occurs when using a minimal amount of heat flux and a Reynolds number with a 2% volume fraction of the TiO_2 nanofluid.

For the purpose improving thermal transfer in the cooling process in electronic devices, TiO_2 -water nanofluids in heat sinks with mini-sized channels were explored by Nakharintr and Naphon [47]. Three different concentrations (0.005%, 0.010%, and 0.015%) of TiO_2 nanoparticles and deionized water were utilized in the synthesis of nanofluids as cooling liquids. The conclusions showed that both nanofluids' thermophysical properties and motion direction in the magnetic field have the potential to impact migration. The thermophysical properties, motion direction, and migration of nanofluids can all significantly impact the heat transfer process.

Maria [31] employed an industrial full-cone pneumatic nozzle to investigate the impact of water-based TiO_2 nanofluids on spray characteristics and thermal conduction and found that TiO_2 nanoparticles did not alter the spray characteristics substantially, but the influence of adsorbed nanoparticles to heat transfer enhancement decreased in the non-boiling regime.

CuO nanoparticles are a commonly studied nanosized metal oxide material. Chein and Chuang conducted a series of experiments to examine the impact of a CuO -water nanofluid as a coolant [48]. The volume fraction of CuO particles in the nanofluids ranged from 0.2% to 0.4%. Their findings indicated that, at low flow rates, a CuO - H_2O nanofluid can remove more heat than pure water in microchannel heat sinks. However, at high flow rates, the nanofluid has minimal effect on additional heat transfer, with heat transfer being primarily influenced by volume flow rate.

Sarafraz also investigated the performance of a CuO - H_2O nanofluid in a heat sink [32], using a CuO - H_2O nanofluid, gallium liquid metal, and water to dissipate the CPU heat under three different states (standby, normal operation, and overload). When compared to the other two working fluids, it was found that CuO - H_2O nanofluid offers a higher thermal performance than water and a lower pressure drop and pumping power than gallium liquid metal when the heat flux is not very high.

The thermal efficiency can be enhanced and the thermal resistance reduced by using various nanofluids. The study conducted by Guo et al. examined the thermal and hydraulic characteristics of ZnO - H_2O nanofluids in a heat sink with micro fins [49]. They used two different models (static and dynamic single-phase models) to study the influence of Brownian motion on the thermal conductivity and rheology. The numerical results show that nanofluids containing ZnO nanoparticles of smaller size and a higher ZnO particle volume fraction perform better than the reverse side. Compared with the static single-phase model, the thermal energy transfer and loss in fluid pressure calculated by dynamic single-phase model are both higher.

2.3. Non-Metallic Nanoparticles

Various non-metallic oxide nanoparticles can be used in data center cooling. Non-metallic oxide and carbon materials are typical non-metallic materials that have been investigated by researchers. SiO_2 - H_2O nanofluids with three differentiated volume frac-

tions (0.2, 0.4, and 0.6) were investigated by Duangthongsuk et al. [50] and the findings contrasted with the outcomes with ZnO–H₂O nanofluids. The findings indicated that various types of particles have varying impacts on the thermal performance of a small circular pin fin heat sink, but they yielded similar results in terms of pressure drop and pumping power. ZnO–water nanofluids demonstrated a 3–9% higher thermal performance compared to SiO₂–water nanofluids.

Carbon materials also can be used in data center cooling. The heat transfer characteristics of nanofluids containing carbon nanotubes (CNTs) at low temperature were studied by Mare et al. [51]. Both the convective coefficient and viscosity of CNT nanofluids were examined at low temperatures from (0 to 10 °C) to explore the convective heat transfer and changes in pressure due to fluid flow. The findings indicate that under laminar flow conditions, CNT nanofluids can enhance the convective heat transfer coefficient by approximately 50% in comparison to pure water at equivalent Reynolds numbers. Additionally, the study highlights the significance of viscosity and pressure drop in the selection of nanofluids for low temperature applications.

Sarafraz and Hormozi [52] utilized multi-walled carbon nanotube (MWCNT)–water nanofluids in a plate heat exchanger to investigate the impact of fouling formation, MWCNT concentration, flow rate, and inlet temperature on the overall heat transfer coefficient and pressure drop. The findings indicated that MWCNT–water nanofluids exhibit a higher friction factor and pressure drop compared to pure water, but the overall thermal performance can obviously be improved after adding MWCNT nanoparticles. With the rise in the volume fraction of the MWCNT, significant fouling resistance can be amplified after long-time operation.

Multi-walled carbon nanotubes (MWCNTs) and graphene nanoplates (GNPs) were investigated by Kumar et al. [53]; samples of 0.5 to 2.0% volume fraction were mixed into water and thermophysical properties including thermal conductivity, viscosity, specific heat, and density were measured. The experimental data divulged that when the volume concentration of the nanofluid was 0.57%, the overall heat transfer coefficient reached its peak value for both the MWCNT and GNP nanofluids. When the concentration continued to increase, the viscosity would increase up to a high level and the pressure drop would be very high. The overall heat transfer coefficient of the MWCNT exceeded that of other nanofluids by a significant margin, standing at 0.75%.

The thermal and hydrodynamic performances of diamond–water and Al₂O₃–water nanofluids in various forms of heat sinks were evaluated by Hasan [54]. Through a numerical method, they modeled the flow and heat transfer of two different nanofluids in a microchannel heat sink, considering three different fin geometries (square, triangular, and circular) as well as those without fins. It was discovered that thermal efficiency can be improved by diamond–water and Al₂O₃–water nanofluids, although this also leads to an increase in pressure drop. Additionally, it was observed that circular fins exhibit the highest heat transfer rate.

Walid et al. investigated the heat transfer of water-based graphene nanofluids under the combined effect of tree-shaped obstacles and an external magnetic field [55]. They compared the flow and heat transfer of graphite nanoparticles with different concentrations under different magnetic fields and fin sizes. The innovation of this study lies in the combination of magnetic fields and tree-shaped obstacles to control the heat transfer and flow of graphene water nanofluids, which can be applied to electronic cooling and thermal management systems.

Shchegolkov et al. further investigated the effect of multi-layer carbon nanotubes and graphite on the distribution homogeneity of the temperature field [56]. The results indicate that the temperature field distribution of the MWCNTs/graphite materials becomes more uniform after the mechanical activation stage.

2.4. Magnetic Nanoparticles

Some magnetic materials, such as Fe_3O_4 nanoparticles dispersed in a non-magnetic base fluid, are called magnetic nanofluids or ferrofluids. In addition to the improvement of heat transfer, as with the other nanofluids, they have both the flowability of ordinary liquids and magnetic features of magnetic materials at the same time. The ferromagnetism they have make it potentially promising to control the flow, motion, heat, and mass transfer of the fluids in external magnetic fields. Some outlying investigations on the utilization of magnetic nanofluids in data centers and electric devices have also been conducted by many researchers.

Magnetic field and magnetic nanofluids were applied in a Pyrex heat pipe and pulsating heat pipes by Gandomkar et al. [57] The effective factors of thermal conductivity characteristics of nanofluids with magnetic properties were tested in heat pipes composed of two different materials (copper and glass). Magnetic nanoparticles were found to improve heat transfer efficiency in pulsating heat pipes and the glass pulsating heat pipes can ensure better stability than those made of copper.

A study of magnetic nanofluids flowing in a micro pulsating heat pipe was conducted by Jahani et al. [58] Their research proved that magnetic nanofluids flowing in a micro pulsating heat pipe are a promising technology in the field of microelectromechanical systems (MEMS). The effect of various working fluids such as water, Ag–water and magnetic nanofluids, heating power ranging from 4 W to 28 W, ratio of charging (20, 40, 60, and 80%), inclination angle, and the utilization of external magnetic field were studied systematically and completely. The findings indicate that the presence of a magnetic field can lead to a decrease in thermal resistance when magnetic nanofluids are flowing. The study can provide an advanced technology for the thermal management of data centers and electronics.

An external magnetic field was used and the wetting characteristics of magnetic nanofluid droplets in a stationary state were investigated by Yu-Chin Chien and Huei Chu Weng [59]. The effects of pore size and magnetic field gradient on the magnetic wetting of magnetic nanofluids on the anodic aluminum oxide surface were studied by an optical test system. Their experimental results showed that with the increase in the pore size of the anodic aluminum oxide, the contact angle also increased. Additionally, the application of a magnetic field gradient can lead to a decrease in the contact angle. Typically, magnetic fields have a greater impact on smaller anodic aluminum oxide pore sizes. They conducted research on the impact of magnetic nanofluid droplets on an aluminum plate's surface when exposed to a magnetic field. The results from the static contact angle indicate that the treatment resulted in a more hydrophilic surface. Observations also revealed that without a magnetic field, the treated surface displayed slightly increased adhesion between the liquid and solid. Additionally, it was noted that under the influence of an external magnetic field, the AAO surface topography could have a significant impact [60]. A comparison of the heat transfer enhancement of different types of nanofluids from the reviewed studies can be seen in Table 1.

Table 1. Comparison of heat transfer enhancement of different types of nanofluids from reviewed studies.

Type of Nanoparticles	Nanoparticles	Base Fluids	Author/Reference	Results
Non-metallic oxide	SiO_2	H_2O	Duangthongsuk [50]	Nanofluid cooling allows a 4–14% enhancement in heat transfer coefficient
Carbon materials	carbon nanotubes (CNT)	H_2O	Mare et al. [51]	Convective heat transfer coefficient enhanced by about 50%

Table 1. Cont.

Type of Nanoparticles	Nanoparticles	Base Fluids	Author/Reference	Results
Carbon materials	Multi-walled carbon nanotube (MWCNT)	H ₂ O	Sarafraz and Hormozi [52]	MWCNT can enhance the thermal conductivity coefficient up to 68%
Carbon materials	Multi-walled carbon nanotube/graphene nanoplate	H ₂ O	Kumar et al. [53]	MWCNT improved relative thermal conductivity by 11.42–22.67%
Carbon materials	Diamond	H ₂ O	Hasan [54]	Diamond–water improved the heat transfer rate by 9.12–9.9%
Metal	Copper	ethylene glycol	Leong [36]	Heat transfer rate showed a 45.2% enhancement
Metal	TiO ₂ /copper	H ₂ O	Wadd et al. [38]	Heat transfer coefficient enhancement of 10–15%
Metal	Copper	ethylene glycol	Sheikhzadeh et al. [39]	Heat transfer rate enhanced by 29.6%
Metal	Ag	H ₂ O	Behrangzade and Heyhat [40]	Overall heat transfer coefficient improved by 16.79%
Metal	Au	H ₂ O	Tsai et al. [41]	Thermal resistance of heat pipe reduced by 20–25%
Metal oxide	Al ₂ O ₃	H ₂ O	Hassani et al. [42]	Performance index of improved by 14.7% and 28.3%
Metal oxide	Al ₂ O ₃	H ₂ O	Nguyen [43]	Heat transfer coefficient enhanced by 40%
Metal oxide	Al ₂ O ₃	H ₂ O	Nnanna [44]	Effective thermal conductivity enhanced by more than 30%,
Metal oxide	TiO ₂	H ₂ O	Ambreen and Kim [45]	26%, 44%, and 62% Raises in Nusselt numbers
Metal oxide	TiO ₂	H ₂ O	Nitiapiruk [46]	Heat transfer coefficient enhanced by over 500 W/m ² ·K
Metal oxide	TiO ₂	H ₂ O	Nakharintr and Naphon [47]	Heat transfer coefficient improved by 12.5%
Metal oxide	TiO ₂	H ₂ O	Maria [31]	TiO ₂ nanoparticles did not alter the spray characteristics substantially
Metal oxide	CuO	H ₂ O	Chein [48]	Thermal conductivity improved by 1.8–8%
Metal oxide	CuO	H ₂ O	Sarafraz [32]	Convective heat transfer coefficient did not surpass that of the gallium
Metal oxide	ZnO	H ₂ O	Guo et al. [49]	Heat transfer coefficient improved by 25.6–38.3%

3. Experimental Studies of Liquid Cooling with Nanofluids

The utilization of nanofluids in liquid cooling mainly focuses on studies of their thermal and rheological properties, as well as their application in microchannel heat sink

heat exchangers. Recent research has focused on the application of nanofluids in phase change heat transfer.

3.1. Thermophysical Properties and Rheological Properties of Nanofluids

Adding nanoparticles to the base fluids can alter the thermodynamic characteristics (such as density, heat capacity, specific heat, and thermal conductivity) and rheological parameters (such as viscosity and shear stress) of the base fluids [61]. The research of Pavia et al. [62] indicates that the thermal conductivity of the base fluids can be greatly enhanced by incorporating nanoparticles. The investigation conducted by Xie et al. [63] further found that by incorporating nanoparticles into the base fluids, it is possible to significantly enhance thermal conductivity. The potential application of nanofluids in heat transfer is enormous due to their improved thermal conductivity. In their study, Keblinski et al. [64] observed that the use of nanofluids in cooling systems resulted in higher heat transfer efficiency and reduced energy input to achieve a specific temperature change.

Luo et al. selected SiC nanoparticles based on insulating silicon carbide oil to synthesise a novel cooling liquid for data center immersion cooling [65]. To compare the heat transfer effect of different coolants, it is essential to compare the values of the heat flow. Heat flow depends on thermophysical properties and rheological characteristics like density, kinematic viscosity, dynamic viscosity, thermal conductivity, and specific heat capacity. Specifically, the heat transfer effect is directly related to the density, thermal conductivity, and specific heat capacity, while it is inversely related to dynamic viscosity. The thermophysical characteristics of nanofluids containing SiC in mineral oil were analyzed, and the effectiveness of using these nanofluids in a cooling system was compared to that of a traditional white mineral oil coolant. Table 2 presents the relevant thermophysical properties of the nanofluids measured by Luo et al. [66].

Table 2. Thermophysical properties of SiC, mineral oil, and 0.3 vol% nanofluid (298.15 K). Reprinted from [66], Copyright (2022), with permission from Elsevier.

Material	Thermal Conductivity (W/M·K)	Dynamic Viscosity (mPa·s)	Specific Heat (J/kg·K)
SiC	165	-	422.0
Mineral oil	0.13	5.6	2197.8
Nanofluid	0.135	5.8	2009.8

They found that the rise in both volume concentration and temperature can lead to the increase in thermal conductivity [67]. By adding 0.3 vol% to 3.7 vol% SiC nanoparticles into base fluids, the thermal conductivity can be enhanced by up to 11.4% and 11.7% compared to the base liquid, respectively. Furthermore, at low Reynolds numbers, the 0.3 vol% nanofluids exhibit better thermal diffusion effects, while at high Reynolds numbers, 3.7 vol% nanofluids exhibit better performance. Their conclusion was that nanofluids based on silicon carbide oil showed great potential for use in immersion cooling for data centers.

Shubhankar et al. investigated the effect of volume fraction of nanoparticles on the heat transfer and flow characteristics of nanofluids [68]. They found that an increase in the volume fraction of silver nanoparticles would lead to an improvement in the thermal conductivity of the nanofluid. This is mainly because the increase in nanoparticles reduces condensation around the particles, thereby improving their relative thermal conductivity. However, an increase in the volume fraction of nanoparticles will lead to an increase in energy consumption, resulting in an increase in the viscosity of the nanofluid. They further found that the larger the volume fraction of nanoparticles, the greater the specific heat of the nanofluid.

According to the research of Al-Rashed et al. [28] and Ambreen and Kim [69], the main reason for the high thermal conductivity of nanofluids is that the thermal conductivity of solid nanoparticles is higher than that of liquids, as well as the space convection between

solid particles and liquids. Its mechanism is that Brownian motion can cause convection, which can elevate the fretting energy in order to improve the convection of energy between particles and fluids. It is the high thermal conductivity of metal nanoparticles that has led to their rapid development in the field of enhanced heat transfer in recent years. In summary, the enhancement of nanofluids' thermal conductivity primarily relies on the thermal conductivity of the nanoparticles.

Another key thermophysical parameter of nanofluids, viscosity, is also related to Brownian motion. The increase in temperature will increase the molecular kinetic energy, leading the Brownian motion effect to be more active, ultimately reducing the intermolecular friction and the dynamic viscosity of the nanofluids.

The Brownian motion effect is related to the size of nanoparticles; the smaller the nanoparticle's size, the stronger the Brownian motion effect will be. Because Brownian motion is generated by the random motion collision of nanoparticles suspended in a base fluid. The smaller the nanoparticle size, the smaller number of liquid molecules that have an impact on the nanoparticle surface. At this point, the imbalance between the particles and the liquid increases, and the nanoparticles accelerate their motion under the force of liquid molecules. On the other hand, when the volume concentration is the same, the smaller the particle diameter, the higher the specific surface area, and the more particles can carry the surrounding fluid in random motion [51].

Studies by Wei et al. [70] and Li et al. [71] have shown that as the volume fraction increases, the viscosity of the nanofluids also increases, and a higher concentration will have lead to a greater impact of temperature on viscosity [72]. The experimental results of Luo et al. [65] show that when the concentration of SiC nanofluid is 10.3%, the viscosity of the nanofluid increases by 35% compared to mineral oil and when the concentration of the SiC nanofluid is 1.7%, the nanofluid experiences only a 1.7% increase in viscosity.

At the molecular level, whether nanofluids can be considered Newtonian fluids mainly depends on the rheological properties of their molecular chains under shear stress. If molecules in a nanofluid are arranged along high-tension planes, its viscosity gradient will decrease and its fluidity will improve with an increase in shear rate. At this point, the nanofluid can be considered a shear-thinning fluid [73]. On the contrary, if the viscosity of the nanofluid increases with an increase in shear stress, the nanofluid is a shear-thickening fluid [74]. The viscosity gradient thus increases under shear stress. Different non-Newtonian fluids have different responses to shear stress.

In our previous work [75], we performed an experiment to investigate how nanofluids impact the thermal conductivity and rheological characteristics of a coolant used in data centers using 0.01–0.15% CuO and Al₂O₃ nanoparticles. The nanoparticles were dispersed into a commercial data center coolant (a typical mineral oil), then the thermal conductivity and rheological properties were measured. Our findings indicated that the addition of a small quantity of nanoparticles can lead to a substantial enhancement in thermal conductivity. The average rise in thermal conductivity was around 20–25%, with no noticeable alterations in the rheological properties of nanofluids compared to the base fluid.

The experimental results of the nanofluid rheological properties measured by Madhusree Kole and T.K. Dey [76] also showed a trend consistent with ours [75]. They found the rheological properties of these mineral oil coolant-based nanofluids to be strongly dependent on the temperature; due to the rise in temperature, there was a significant decrease in viscosity following a non-linear pattern.

3.2. Nanofluids as Coolant in Microchannel Heat Sinks

The most widely used apparatus for electric devices and data center cooling is the microchannel heat sink. The application of nanofluids in improving heat transfer in microchannel heat sinks (MCHSs) has been extensively studied for over 20 years. Despite the promising findings in some studies, conflicting results and a lack of understanding of the mechanisms behind heat transfer enhancement have impeded further advancements in utilizing nanofluids as coolants [77].

Li et al. [78] conducted research on the heat dissipation capability of plate fin vapor chamber heat sinks using infrared thermography to analyze the impact of varying fin width, height, and quantity. The effect of the Reynolds number on heat transfer efficiency is also investigated in their experiment. The results were compared with the conventional heat sinks made of aluminum. The results show that heat transfer in a vapor chamber heat sink is distributed more evenly compared to a traditional aluminum heat sink, so the heat can be taken away faster and a reduction in temperature can be effectively achieved. As the Reynolds number increases, the overall thermal resistance decreases and when the Reynolds number is low, fin dimensions have a more obvious effect on the heat transfer.

Sui et al. investigated the heat transfer performance and pressure drop of wavy channels using numerical and experimental methods [79]. They found that when the Reynolds number was between 300 and 800, wavy channels had significant advantages compared to traditional straight channels. The research results of Gong et al. indicate that when the Reynolds number exceeds 50, the heat transfer and flow performance of the wavy channel are improved by 55% compared to straight channels [80]. Lu et al. found that using porous fins in wavy microchannels can effectively reduce flow thermal resistance and decrease flow pressure drop [81]. They attributed the reason for the decrease in pressure drop to the permeation and slip effects exhibited by the coolant in this new structure. Netami et al. used a multi-objective genetic algorithm and found that the slight disturbance caused by the corrugated channel can reduce the overall thermal resistance by 87%, while the pumping power only increased by 10% [82].

Thus, in order to achieve an optimal performance from vapor chamber heat sinks, the number of fins must be optimized. They concluded that vapor chamber heat sinks have better heat transfer performance. Yang et al. [83] investigated four different types of plain-fin heat sinks including plate fins, interrupted fin shapes, and both dense and sparse vortex generators. Besides the effect of different fin types on heat transfer, the pressure drop caused by the addition of fins was also considered. They found that at a 3–5 m/s frontal velocity, the surface area can be reduced by 12–15% with a triangular attack vortex generator, so a triangular attack vortex generator is an optimal design.

Plate fin and pin fin heat sinks were investigated by Kim et al. [84] through experiments and analytical methods. Different flow rates and channel widths were studied and jet impingement was also considered. A model based on their experimental data was developed to forecast the thermal resistance of heat sinks and the decrease in pressure as fluid flows through them. Furthermore, they suggested a contour map which can be used to compare the thermal resistances associated with plate fin and pin fin heat sinks at different dimensionless pumping powers. The thermal resistance ratio of the heat sinks in the contour map can be determined based on the dimensionless length and pumping power.

Li et al. [85] conducted a study on the thermal characteristics of coolant flowing in plate fin heat sinks and how they are affected by geometrical parameters such as groove width, depth and pitch, fin dimensions, and Reynolds number. They carried out both experimental and numerical studies. They found that raising the impinging Reynolds number can improve heat transfer. As the fin width increased, there was a rapid decrease in thermal resistance at first, but the thermal resistance increased rapidly after the fin width surpassed an optimum value. Both an impinging Reynolds number and fin width have an optimum value during the process of increasing the fin height to obtain a low thermal resistance.

Research on nanofluid-based heat dissipation in trapezoidal grooved microchannel heat sinks was conducted by Kuppusamy et al. [86]. Numerical methods were used to investigate various geometrical parameters, and types and concentrations of nanofluids. Particle diameter, base fluid, and Reynolds numbers were also studied. They used the finite volume method to address the governing energy equations of 3D laminar flow and conjugate heat transfer. They found that expanding the maximum width and narrowing the minimum width of the trapezoidal groove can enhance thermal efficiency. This suggested that triangular shape microchannel heat sink performed better than the rectangular

shape. They also gave the best volume fraction and particle diameter of water-based Al_2O_3 nanofluids.

Microchannel heat sinks with $\text{Cu-H}_2\text{O}$ and diamond- H_2O nanofluids were studied by Jang and Choi [87]. The volume fraction they used in their research is 1% and the sizes of Cu and diamond nanoparticles are 6 nm and 2 nm, respectively. The findings indicate that the use of nanofluids can lead to a 10% enhancement in heat transfer performance. It was ultimately concluded that the pairing of nanofluids with microchannel heat sinks has the capability to effectively reduce ultra-high heat flux by up to 1350 W/cm^2 .

A similar conclusion was also achieved by Jung et al. [88]. They examined the convective repositioning of thermal energy and movement of liquids with 170 nm Al_2O_3 nanofluids in microchannels. The convective heat transfer coefficient can be enhanced by up to 32% with the existence of 1.8% volume fraction Al_2O_3 nanofluids at the same time, and no obvious friction loss was observed.

Lim et al. [89] conducted a study on the heat transfer and fluid flow in a triangular grooved micro-channel heat sink using various types of particles, different volume fractions, and base fluids. They also examined the impact of groove geometrical parameters such as angle, depth, and pitch on heat transfer. The findings indicated that 0.04% 25 nm $\text{Al}_2\text{O}_3\text{-H}_2\text{O}$ can significantly improve heat transfer efficiency by up to 179.55% compared to traditional microchannels.

The effect of the location of the heater and the effect of nanofluids in a microchannel heat sink were studied by Anbumeenakshi and Thansekhar under a non-uniform heating condition [90]. They used three separate heaters in their experiment and could switch the status of any two heaters so that different non-uniform heating conditions could be created. The results showed that the position of heater can very much influence the hotspot of the heat sink.

Sivakuma et al. [91] investigated the heat transfer in a microchannel heat sink with a serpentine shape under forced convection. The coolants they used were $\text{Al}_2\text{O}_3\text{-H}_2\text{O}$ and $\text{CuO-H}_2\text{O}$ nanofluids. The experimental values were also compared to the convective heat transfer coefficient calculated from theoretical correlations. Their experiments showed that nanofluids can obviously enhance the convective heat transfer coefficient of pure water, and compared with $\text{Al}_2\text{O}_3\text{-H}_2\text{O}$ nanofluids, $\text{CuO-H}_2\text{O}$ had a more effective convective heat transfer. They also found that larger nanoparticles' fraction led to a larger forced convective heat transfer coefficient.

Sakanova et al. [92] studied the passive heat transfer of nanofluids in a wavy channel. They used diamond- H_2O , $\text{SiO}_2\text{-H}_2\text{O}$, and $\text{CuO-H}_2\text{O}$ nanofluids as coolants and studied their heat transfer and fluid movement in wavy channels. They studied the influence of the wavy channel on thermal and flow thermal resistance, pressure drop, and friction factor.

Selvakumar and Suresh applied $\text{CuO-H}_2\text{O}$ nanofluids to a thin-channel copper water block (see Figure 5) to study the interface temperature and convective heat transfer coefficient of the water block, and compared the result to deionized water [93]. They found that $\text{CuO/H}_2\text{O}$ nanofluids can bring as much as a 29.63% enhancement to the convective heat transfer coefficient. Nevertheless, there was a 15% rise in the pumping power for nanofluids.

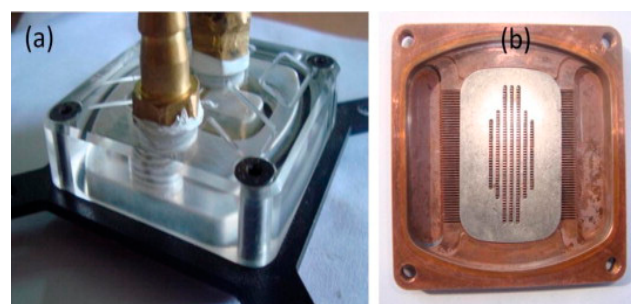


Figure 5. (a) Image of water block; (b) copper base with a thin-channel design and jet plate. Reprinted from [93], Copyright (2012), with permission from Elsevier.

As shown in Figure 6, a heat sink made of ERG aluminum foam designed for the Intel core i7 CPU was studied by Bayomy and Saghir [94]. They studied the heat transfer characteristics and thermal performance of γ - Al_2O_3 -water nanofluid. The Reynolds numbers in their experiment covered the entire non-Darcy flow regime. The results showed that the increase in the nanofluid's fraction had a positive effect.

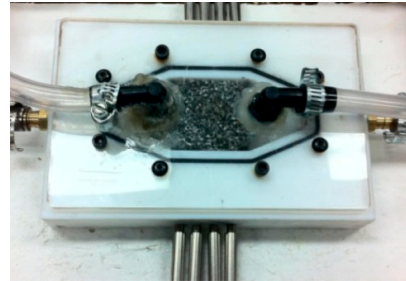


Figure 6. The aluminum foam heat sink used to remove heat from Intel core i7 CPU. Reprinted from [94], Copyright (2017), with permission from Elsevier.

A vapor compression refrigeration system (VCRS) is shown in Figure 7. Jeng and Teng developed a Al_2O_3 -water nanofluid liquid cooling system instead of using distilled water [95]. Their experiment showed that the nanofluid performed better in terms of heat releasing and temperature on the surface of the heater, but the water pump of nanofluids consumed more power.

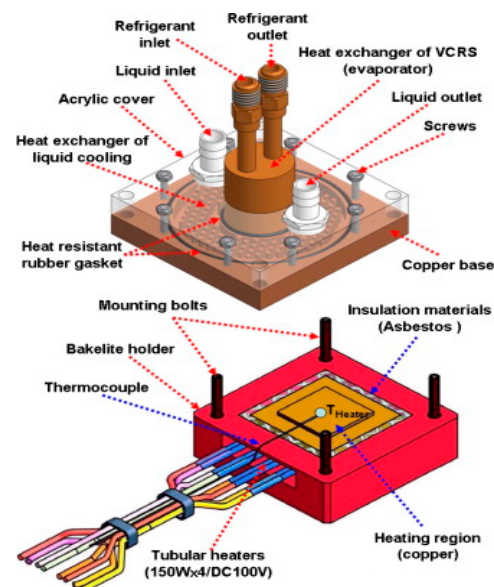


Figure 7. Schematic picture of the combined heat exchanger and heating module structure. Reprinted from [95], Copyright (2013), with permission from Elsevier.

In order to improve the cooling performance of conventional coolant, Murshed et al. [96] conducted a study using a mini channel heat sink made of Cu attached to an electronic heat source. The study focused on the cooling effectiveness of a Al_2O_3 - H_2O nanofluid, and the findings indicated that it could serve as a superior coolant for electronic cooling applications.

3.3. Phase Change Heat Transfer of Nanofluids

For high-power density servers, traditional single-phase liquid cooling cannot provide sufficient cooling capacity, therefore phase change cooling is required [97]. Single phase immersion cooling is the process of immersing a server in a sealed container and removing

heat through the coolant in the container. During the heat transfer process, the coolant remains in the liquid phase. The heated coolant will enter the heat exchanger for cooling and then be recycled back to the server [98]. On the other hand, two-phase immersion cooling takes away the heat generated by high-power density servers through the boiling phase transition of the coolant [99]. The coolant absorbs waste heat from the high-power density server through evaporation and then condenses in the heat exchanger. Due to the latent heat generated during the evaporation and condensation processes, the two-phase immersion cooling releases more heat. Due to its efficient heat dissipation mechanism and high heat transfer coefficient, two-phase immersion cooling can provide efficient and energy-saving cooling for high-power density electronic devices compared to single-phase immersion cooling [100]. Microchannel flow boiling heat transfer technology, which has outstanding thermal conductivity and economical production expenses, has recently attracted the interest and wide application of researchers [101,102].

They synthesized Al_2O_3 nanofluids for refrigerant boiling heat transfer in a flowing system using microfluidic flow synthesis method, and specifically designed a spiral microreactor for Al_2O_3 nanofluids for phase change heat transfer [103]. Their schematic diagram is shown in Figure 8. Their fluorescence experiment results showed that the spiral structure has excellent mixing efficiency. In addition, the size, purity, crystal form, and stability of the Al_2O_3 nanofluid were also characterized. Finally, the device shown in Figure 8 was used for the flow boiling experiments of the Al_2O_3 nanofluids with different concentrations. By analyzing the wettability, deposition, and visualization images, the mechanism by which Al_2O_3 nanofluids enhance flow boiling heat transfer was revealed.

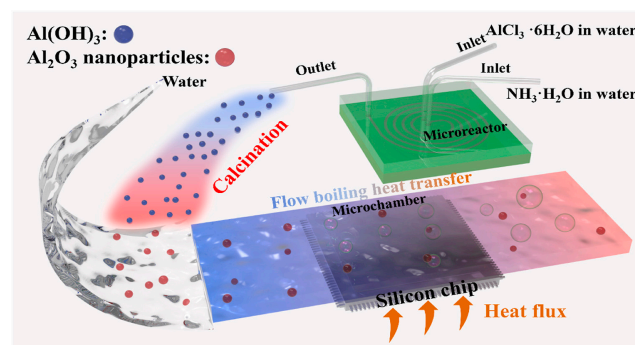


Figure 8. Schematic representation of flow boiling in alumina nanofluid using a spiral microreactor. Reprinted from [103], Copyright (2024), open access article permits unrestricted use.

4. Simulations and Theoretical Studies of Liquid Cooling with Nanofluids

Suja et al. applied Ansys Fluent 2020R1 (a commercial CFD code) to study the thermal performance of water–ethylene glycol mixture-based nanofluids in porous square pin fin heat sinks and the numerical results were compared with experimental values [104]. The grid structure of the computational domain can be seen from Figure 9. Three kinds of nanoparticles (CuO , Al_2O_3 , and TiO_2) and two volume fractions (0.5% and 2%) were used in their study. The pin fins of the heat sinks are arranged in three ways; inline, staggered, and 45° inline, and the effects of three different porosity values (40%, 50%, and 60%) on the thermal and fluidic behavior were studied. The findings indicated that the inclusion of nanoparticles leads to a notable enhancement in the cooling effect, among which CuO –water nanofluid had the best heat transfer performance. When the square needle fin heat sinks adopted a 45° inline configuration, the heat transfer effect was the best. Their simulation results provide valuable insights for high-performance cooling systems for electronic devices.

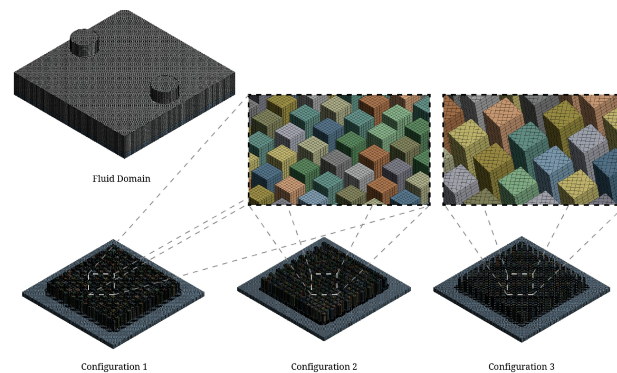


Figure 9. Grid structure of the computational domain. Reprinted from [104], Copyright (2024), open access article permits unrestricted use.

Heidarshenas et al. conducted numerical simulations to investigate the cooling effect of water-based Al_2O_3 nanofluids in a pin fin heatsink [105]. This study altered the geometric structure of the pin fin heatsink (distance, height, and diameter of circular pin fins) and modeled the nanofluid flow inside the pin fin heatsink using a two-phase approach, calculating the values of thermal resistance (THR) as well as temperature uniformity (Teta) of the pin fin heatsink, and the heat transfer coefficient (HTC). The research results show that increasing the height of pin fins leads to high Teta and low HTC and heat capacity values. The larger the size of the pin fins and the spacing between them, the higher the HTC and Teta values and the lower the THR. It is worth mentioning that artificial intelligence (AI) techniques were also applied in their study to optimize the obtained results. With the help of the analysis of results through artificial intelligence, a minimum HEK, maximum HTC, and the best Teta can be obtained.

Castillo et al. applied numerical methods to study the thermal and hydraulic performance of shear thinning nanofluids in microchannel heat sinks with five different geometries [106]. They modified the single-phase expression of thermophysical properties in numerical simulations to make it suitable for hybrid nanofluids and used apparent viscosity contour lines to assess the thermal and hydraulic efficiency of nanofluids. Based on their previous experimental data [107], they compared the heat transfer effects of the water–ethylene glycol mixture-based Al_2O_3 and multi-walled carbon nanotube nanofluids at Reynolds numbers of 200–1200 through numerical simulations. They found that an increase in pin density can lead to an enhancement in heat transfer rate and hydraulic loss, as well as a risk of thermal blocking. The research results indicated that the lower the power law index of nanofluids, the better they are with regard to heat removal and loss of pressure. The final conclusion is that fluids with thinning rheological properties are suitable for cooling electronic devices in microchannel heat sink devices.

The influence of a water-based Al_2O_3 nanofluid on a fin channel heatsink was studied by Kavitha et al. using numerical simulation methods [108]. They used the ANSYS-CFX software to study the heat transfer characteristics, resistance to thermal flow, and the Nusselt number of water-based Al_2O_3 nanofluids with different volume concentrations and inlet velocities, then compared the results with that of distilled water. The results showed that under different boundary conditions and at various inlet velocities (0.02 m/s to 0.10 m/s), the thermal resistance of water-based Al_2O_3 nanofluid was much lower than that of pure water, and it can increase the heat transfer coefficient by up to 68%. They concluded that the employment of Al_2O_3 –water nanofluids in finned heat sinks with channels can significantly reduce the temperature on the outer layer and the energy consumption of electronic chips, thereby improving cooling performance compared with traditional methods. Due to the high thermal conductivity, low precipitation rate, the most reliable emulsification, and the stability of Al_2O_3 nanoparticles, as well as the wide application of channel finned heat sinks with fins in the cooling of tiny electronic components, their study has broad reference value for nanofluid cooling.

To forecast the thermal conductive properties of nanofluids, many mathematical models can be applied. In our previous work, the Maxwell model [67,109] has been proposed for the prediction of thermal conductivity in nanofluids, and is represented by Equation (1) below.

$$\frac{k_{nf}}{k_{bf}} = \frac{2k_{bf} + k_{np} + 2\varphi(k_{np} - k_{bf})}{2k_{bf} + k_{np} - \varphi(k_{np} - k_{bf})} \quad (1)$$

The effective thermal conductivity of the solid–liquid suspension, denoted as k_{nf} , is determined by the thermal conductivity of the base fluid (k_{bf}), the thermal conductivity of the nanoparticles (k_{np}), and the volume fraction of the nanoparticles (φ).

In addition to the Maxwell model, the Hamilton and Crosser model [102,110] is also widely applied. Hamilton and Crosser made adjustments to the Maxwell model by incorporating the sphericity of nanoparticles into the equation. The Hamilton and Crosser model is applicable for determining the thermal conductivity of nanofluids created by dispersing nanoparticles of varying shapes in the base fluid. The equation of the Hamilton and Crosser model is as follows:

$$\frac{k_{nf}}{k_{bf}} = \frac{k_{np} + (n-1)k_{bf} + (n-1)(k_{np} - k_{bf})\varphi}{k_{np} + (n-1)k_{bf} - \varphi(k_{np} - k_{bf})} \quad (2)$$

In Equation (2), n is the shape factor related to sphericity degree. Sphericity degree (Ψ) is defined as the ratio between the surface area of a sphere and a particle at an equal volume. The equation for calculating the shape factor (n) through sphericity degree (Ψ) is as follows:

$$n = \frac{3}{\Psi} \quad (3)$$

By substituting the parameter k for the $(n-1)$ factor, Yamada and Ota [111] modified the Hamilton and Crosser models to the following Equations (4) and (5):

$$\frac{k_{nf}}{k_{bf}} = \frac{k_{np} + kk_{bf} + k(k_{np} - k_{bf})\varphi}{k_{np} + kk_{bf} - \varphi(k_{np} - k_{bf})} \quad (4)$$

$$k = 2\varphi^{-0.2} \quad (5)$$

For cylindrical nanoparticles, the parameter k in the Yamada and Ota model is expressed as the following form:

$$k = 2\varphi^{0.2} \frac{L}{d} \quad (6)$$

where the length and diameter of the nanoparticles are denoted as L and d , respectively.

When there is a need to determine the rate of heat transfer between particles in a solid or the thermal conductivity from nanoparticles to the base fluid, a model proposed by Davis can be taken into consideration. The expression of the Davis model [112] is shown in Equations (7) and (8):

$$\frac{k_{nf}}{k_{bf}} = 1 + \frac{3(1-\alpha)\varphi}{(1+2\alpha) - (1-\alpha)\varphi} [\varphi + f(\alpha)\varphi^2 + O(\varphi^3)] \quad (7)$$

$$f(\alpha) = \sum_{p=6}^{\infty} \left[\frac{B_p - 3A_p}{(p-3)2^{p-3}} \right] \quad (8)$$

In Equation (8), The constants A_p and B_p are associated with the parameters α and p . It should be noted that the Davis model can only be employed to calculate the thermal conductivity of low concentration nanofluids synthesized from spherical nanoparticles.

In summary, as the first proposed model, the Maxwell model is widely applicable for predicting the thermal conductivity of two-phase coolants composed of suspended solid particles and base fluids. The Hamilton and Crosser model can be used to calculate

the thermal conductivity of nanoparticles with different shapes, regardless of whether the nanoparticles are spherical or non-spherical. The Davis model has strict application conditions and can only be used for low volume fraction and spherical nanoparticles.

In our previous published works, various nanoparticles (CuO, Cu₂O, TiO₂, Al₂O₃, etc.) were dispersed in different cooling media, including ethanol [113], methanol [114], and a commercially available data center coolant [75]. We measured the rheological properties of nanofluid coolants and fitted them using different empirical formulas, including Einstein's model [115], Brinkman's model [116], Batchelor's model [117], and Wang et al.'s model [118].

The Einstein model is the first theoretical framework for forecasting the viscosity of mixtures and composites. It operates under the assumption that nanofluids can be represented as linear viscous fluids containing spherical particles. This approach has been extensively validated for its efficacy when the volume concentration (φ) of nanofluids is below 0.02. Equation (9) is the expression of the Einstein model [115]:

$$\mu_{nf} = \mu_{bf}(1 + 2.5\varphi) \quad (9)$$

The Brinkman model [116] is a modified version of the Einstein model, which expands its applicability to a volume fraction of up to 0.04.

$$\mu_{nf} = \mu_{bf}(1 - \varphi)^{-2.5} \quad (10)$$

Batchelor [117] considered the influence of Brownian motion within the model shown in Equation (11):

$$\mu_{nf} = \mu_{bf}(1 + 2.5\varphi + 6.5\varphi^2) \quad (11)$$

Wang et al.'s equation [118] is listed below:

$$\mu_{nf} = \mu_{bf}(1 + 7.3\varphi + 123\varphi^2) \quad (12)$$

For all the above equations, μ_{nf} stands for the viscosity of nanofluids while μ_{bf} stands for the viscosity of base fluids, and φ represents the volume concentration of nanoparticles.

By using the models above, empirical correlations were derived, thus quantitatively valuable insights into the nanofluids system can be obtained.

5. The Challenges of Nanofluids Application in Data Centers

5.1. Stability of Nanofluids

The incorporation of nanoparticles into the primary fluid can greatly improve the efficiency of heat transfer, but it is not easy for nanoparticles to remain stable in the base fluid [119]. Unstable nanoparticles tend to agglomerate, aggregate, and form clusters, causing nanoparticle separation from the base fluid and the sedimentation at the bases. If the problem of the stability of nanofluids can be solved, nanofluids will have a significant impact on improving heat transfer efficiency.

At present, the main theory revealing the mechanism behind agglomeration is the theory of electrostatic stabilization [119,120]. According to this theory, if the Van der Waals attraction force between different nanoparticles exceeds electric double layer repulsive force (EDRF), the collision of nanoparticles can occur [121]. During the collision, the nanoparticles will agglomerate and form clusters. Finally, they can settle because of the effect of gravity, which will lead to the deterioration of the thermal and rheological properties of the nanofluid, and cause blockage in microchannels. Contrarily, if the electric double-layer repulsive force exceeds the Van der Waals force, the nanoparticles will separate with each other and agglomeration can be avoided.

To verify the stability of nanofluids, it is not only essential to characterize the nanoparticles, but also to characterize the parameters of the synthesized nanofluids.

The basic parameters of nanoparticles can be characterized through many different techniques. In order to obtain the morphology of nanoparticles and visually observe their

microstructure, shape and size, scanning electron microscopy (SEM) and transmission electron microscopy (TEM) can be applied [122,123].

Atomic force microscopy (AFM) can be applied to obtain the surface topography [124]. The crystalline structure of the nanoparticles can be obtained through X-ray powder diffraction (XRD) [125]. Fourier transform infrared spectroscopy (FTIR) can be used to examine surface chemistry of solid nanoparticles [126]. Mass loss of specimen and variations in physical characteristics as temperature increases can be measured through thermo-gravimetric analysis (TGA) and differential scanning calorimetry (DSC), respectively [127,128]. Some characterization results of nanofluids can be seen from Table 3.

Table 3. Characterization methods of nanofluids.

Type	Reference	Main Method	Result
P25 TiO ₂	Our previous work [122]	SEM, XRD	The TiO ₂ nanoparticles exhibit a spherical shape with an average size of approximately 21 nm
Copper	Bhat [123]	TEM	TEM images provided visual proof of the development of particles exhibiting spherical shape
Al ₂ O ₃	Peng [124]	AFM, UV–vis	The presence of nanoparticles on the heating surfaces was demonstrated
Al ₂ O ₃ –CuO	Bhavani [125]	XRD, FTIR	The characteristic peaks confirmed the presence of CuO and Al ₂ O ₃
Graphene oxide	Aich [126]	FTIR, DSC	The existence of C=O stretching vibrations and a highly organized honeycomb-like structure in the graphene architecture was denoted
AP–SiO ₂ , β-CD–SiO ₂	Li [127]	TGA	β-CD was successfully loaded on the surface of SiO ₂
SiC	Vallejo [128]	DSC	The isobaric heat capacities for the base fluid and nanofluids were obtained

There are also many methods to assess the stability of nanofluids. A basic inspection method is visual inspection. In this simple method, the synthesized nanofluid is kept in a steady condition and observed at various time intervals. Stability without precipitation indicates that the nanofluid has good stability. Yu [129] and Tang [130] employed visual inspection to prove that water-based MWCNT nanofluids can stay stable for 90 days and 60 days, respectively. Although this method can give intuitive results, it takes a long time.

Another technique to test the stability of nanofluids through observation of sedimentation, which takes less time, is the centrifugal method [21,131]. In this method, nanofluids were filled in a plastic centrifuge and rotated for several minutes. If no sedimentation was observed in the sample, it indicates that the nanofluid has good stability. Transmission electron microscopy (TEM) also can be used to investigate the stability of nanofluids by capturing high-resolution images of particle distribution in the base fluid directly.

In addition to the above methods, the stability of nanofluids can also be quantitatively determined by measuring the zeta potential and mean nanoparticle size through dynamic light scattering (DLS). Zeta potential (ξ) measures the potential difference between the base fluid and the stern layer of the synthesized nanofluid. If the zeta potential (ξ) value is above 60 mV, it means the sample has good stability. Nanofluids with zeta potential (ξ) values between 40 and 60 mV exhibit moderate stability, while a zeta potential (ξ) value below 40mV indicates the bad stability of nanofluids [132].

As one of the widely applied techniques to study the stability of nanofluids, the dynamic light scattering (DLS) method can measure mean nanoparticle size and size distribution in nanofluids [75].

5.2. The Erosion and Abrasion of Nanofluids

Nanofluid contains nanosized solid particles, but most literature on nanofluids concentrates only on the characteristics of nanofluids, ignoring the wear and corrosion of nanoparticles on the surfaces of electronic components during the flow process. During the circulation of nanoparticles following the coolant, nanoparticles will collide with the surface of microchannels. After prolonged operation, the nanoparticles will wear the microchannels. In addition, nanoparticles' conductivity may adversely affect the regular functioning of liquid cooling systems in data centers. Therefore, the erosion and corrosion issues have a substantial influence on the sustained and stable functioning of nanofluids in real industrial applications in the long term.

Stachowiak pointed out the difference between wear by erosion and abrasion [133]. Erosion refers to the damage to the solid surface caused by particle impacting against a solid surface. The wear caused by abrasion means harder particles pass over solid surfaces and lead to material loss. Unlike erosion and abrasion, corrosion refers to the chemical damage on the solid surface, which in turn enhances the chemical stability on the surface of a material [81].

5.3. Electrical Insulation of Nanofluids

Nanoparticles commonly used in nanofluids include metals and metal oxides, such as Cu, Ag, Al_2O_3 , and CuO, which have some electrical conductivity. Based on the requirement for a dielectric coolant, the electrical conductivity (EC) must be below $10^{-9} \mu\text{S}/\text{cm}$ [134]. Only dielectric coolants that meet the standards of insulation coolants can be reliably used for direct cooling applications, such as immersion cooling, spray, and jet cooling. When the coolant exhibits an electrical conductivity exceeding $100 \mu\text{S}/\text{cm}$, it can conduct electricity in direct contact liquid cooling for data centers and can only be used for indirect cooling applications [135].

Khdher investigated the electrical conductivity of nanofluids containing Al_2O_3 nanoparticles in 100% bio glycol (BG) as a base fluid [136]. As shown in Figure 10, the measuring instrument they used was a Cyberscan PC-10 (Eutech Instruments, Singapore), a four-cell conductivity electrode meter with built-in temperature compensation (ATC). The electrode meter they used has the function of measuring instantaneous conductivity. Various volume concentrations (0.1, 0.3, 0.5, 0.7, and 1%) of Al_2O_3 nanofluids were tested from 30 to 80°C and the measurement showed the electrical conductivity of 0.5% Al_2O_3 nanofluids raised from 53 to $154 \mu\text{S}/\text{cm}$ compared to pure bio glycol. Ganguly et al. [137] and Sarojini et al. [138] conducted experimental works separately on water-based Al_2O_3 nanofluids and obtained similar conclusions that electrical conductivity increased by adding nanoparticles.



Figure 10. The instruments used to measure electrical conductivity. Reprinted from [136], Copyright (2016), with permission from Elsevier.

Subramaniyan dispersed TiO₂ nanoparticles into different types of base fluids (water, ethanol, and propanol) and studied the electrical conductivities [139]. The results showed that the dielectric constants were influenced by the volume concentrations of nanofluids and among the three kinds of nanofluids, the dielectric constants of water-based TiO₂ nanofluids were higher than the others.

6. Research Directions in the Future

In response to the challenges faced by nanofluids mentioned above, more efforts can be made in the following directions, as shown in Figure 11.

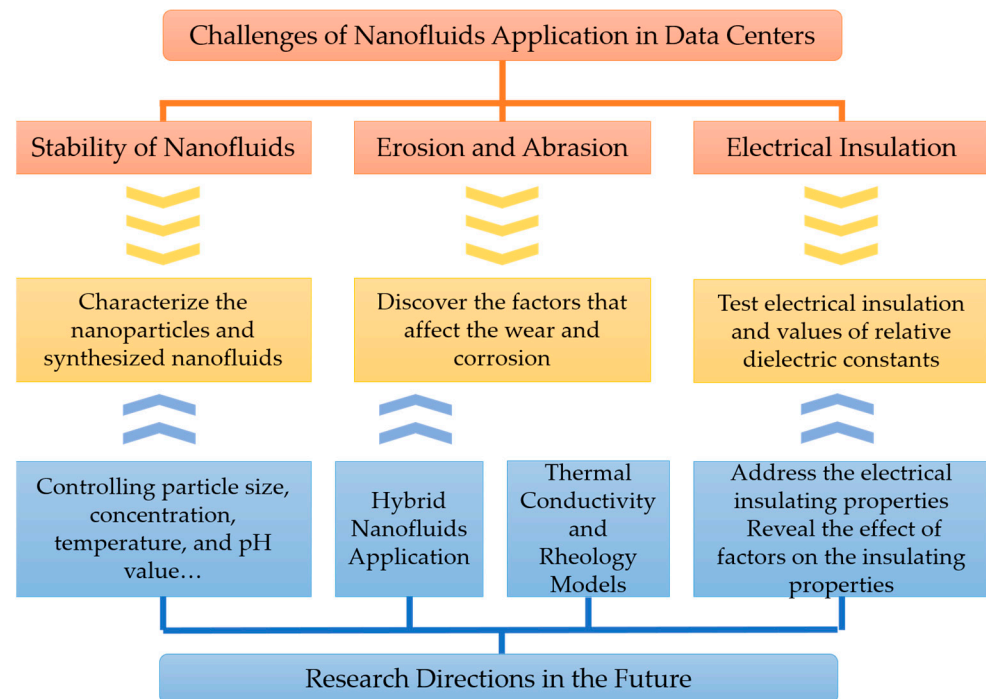


Figure 11. Approaches to improve nanofluids and research directions in the future.

6.1. Nanofluids Stability Improvement

Maintaining the stability of nanosized particles in different base fluids and preventing the agglomeration of nanoparticles are the primary problems in the application of nanofluids. There are numerous elements that have effect on the stability of nanoparticles, including the particle size, concentration, temperature, and pH value, etc. Future research should focus on the influence of these factors and the mechanisms behind them; in particular, the combined effect of these factors, not just a single factor.

6.2. Hybrid Nanofluids Application

The hybrid nanofluids are novel nanofluids synthesized by dispersing various kinds of nanoparticles. Current research has confirmed that hybrid nanofluids can improve thermal conductivity and heat transfer efficiency. However, there is not enough research on hybrid nanofluids for liquid cooling in data centers. The hybrid nanofluids that have been studied include Al₂O₃–Cu nanoparticles [140], graphene nanoplatelets–Ag nanoparticles [141], SiC–CuO–Al₂O₃ nanoparticles [142], etc. These studies confirmed that hybrid nanofluids have promising prospects in the field of liquid cooling in data centers. Therefore, hybrid nanofluids need to attract more attention from researchers in the future.

6.3. Reversal of Erosion and Abrasion in the Application of Nanofluids

In order to achieve long-term safe and stable operation of nanofluids as coolants, wear and corrosion issues cannot be ignored. Numerous experimental investigations are needed

to discover the factors that affect the wear and corrosion of nanoparticles on the surface of electronic devices in order to find ways to avoid or mitigate wear and corrosion and find the best nanofluids.

6.4. Thermal Conductivity Models and Rheology Models

In the current research on nanofluids, several thermal conductivity and rheological models already exist. However, these models have been simplified accordingly, and the accuracy needs to be further improved. Each model is specific to a specific kind of nanofluid, and there is a lack of universally applicable and uniform models. In addition, with the increasing application of hybrid nanofluids, more research should be conducted on thermal conductivity and rheological models for two or more nanofluids.

6.5. Electrical Conductivity Characteristics

Nanofluids applied in the scenario of thermal exchange in data centers require excellent non-conductivity. In studies related to nanofluids, electrical insulation tests should be included and values of relative dielectric constants should be given. More future studies should address the electrical insulating parameters of nanofluids, exposing the impact of these factors on the insulation capabilities.

7. Conclusions and Perspectives

Compared with traditional heat transfer coolants, nanofluids can significantly enhance thermal conductivity and heat transfer efficiency, which has attracted a lot of attention in various heat transfer research fields. On the other hand, more research is needed on the utilization of nanofluids in the cooling technology in data centers that consume a substantial quantity of power for heat dissipation. This review article briefly introduced the liquid cooling technology in data centers, then put forward a detailed overview of the utilization of nanofluids in data center cooling, such as the types of nanofluids, heat dissipation devices using nanofluids, heat transfer and rheological properties, parameters that affect the stability of nanofluids, including experimental research, numerical research, and theoretical models. Finally, we analyzed the factors that hinder the application of nanofluids in data centers and provided suggestions for future research directions. In conclusion, the following remarks can be summarized from this article:

1. Numerous studies have demonstrated that the thermal conductivity of nanofluids can be greatly enhanced by incorporating nanoparticles. These studies confirm the enormous potential of nanofluids as data center coolants. The current studies only focus on the nanofluids themselves, lacking evaluation of their effectiveness in cooling systems;
2. Different types of nanofluids, including metal, metal oxide, non-metallic, and magnetic nanofluids, and various cooling systems, such as microchannel heat sinks, have been analyzed in different categories. There are different techniques to investigate the application of nanofluids in data centers, including experimental research, numerical simulation, and theoretical models;
3. Currently, there is a significant amount of research focused on the thermal and physical characteristics of nanofluids, including their specific heat and thermal conductivity. However, there has been relatively limited investigation into the rheological properties of nanofluids. This article focuses on the analysis of the thermal and physical characteristics as well as the rheological parameters of commonly used nanofluids such as Al_2O_3 , TiO_2 , and CuO and compares them with traditional coolants. Nanofluids can significantly improve thermal conductivity, while some studies suggest that nanofluids do not cause an obvious increase in viscosity;
4. The literature suggests that using nanofluids as coolants in data centers presents some challenges, including the stability of nanofluids, particle erosion and wear on electronic devices and solid surfaces, and the limitations of nanofluid electrical conductivity. Among these, the stability of nanofluids has significant limitations on their use in data center liquid cooling. Hence, it is essential to uncover the factors influencing the

stability of nanofluids, enhance production techniques for nanofluids, and ultimately boost the stability of nanofluids;

5. Finally, concerning the potential areas for future investigation into the application of nanofluids as cooling agents in data centers, our article proposes following five suggestions, including improving their stability, employing hybrid nanofluids, developing more accurate thermal conductivity and rheology models, and decreasing electrical conductivity.

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Nomenclature

Al_2O_3	Aluminum oxide
AgO	Silver oxide
Cu	Cuprum
CuO	Copper oxide
CFD	Computational fluid dynamics
CNT	Carbon nanotubes
EG	Ethylene glycol
Fe_2O_3	Ferric oxide
Fe_3O_4	Ferric oxide
MWCNT	Multiwalled carbon nanotube
SiC	Silicon carbide
SiO_2	Silicon dioxide
SWCNT	Single-walled carbon nanotube
TiO_2	Titanium oxide
Zn	Zinc
ZnO	Zinc oxide

Symbols

k_{nf}	Nanofluid thermal conductivity ($\text{W}/\text{m}\cdot\text{K}$)
k_{bf}	Base fluid thermal conductivity ($\text{W}/\text{m}\cdot\text{K}$)
k_{np}	Nanoparticle thermal conductivity ($\text{W}/\text{m}\cdot\text{K}$)
K	Heat transfer coefficient ($\text{W}/(\text{m}^2\cdot\text{K})$)
L	Nanoparticle length (nm)
d	Nanoparticle diameter (nm)
n	Shape factor
p	Constant
α	Constant
μ_{nf}	Nanofluid viscosity ($\text{Pa}\cdot\text{s}$)
μ_{bf}	Base fluid viscosity ($\text{Pa}\cdot\text{s}$)
ρ	Density (g/cm^3)
φ	Nanofluid volume fraction
Ψ	Sphericity degree

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